

Madhan Lankalapalli

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EDUCATION

Bachelor of Technology – Electronics and Communications Engineering

2022 – 2026

Rajiv Gandhi University of Knowledge Technologies (RGUKT AP IIIT), Nuzvid, Andhra Pradesh, India

CGPA: 8.7 / 10

Relevant Coursework: Digital System Design, VLSI Design Flow, Physical Design, Embedded Systems, Computer Architecture

SKILLS SUMMARY

- HDL / Programming:** Verilog HDL, C++, Python
- FPGA Tools & EDA:** Xilinx Vivado, Vivado HLS, Vitis IDE, Vitis AI, MATLAB, MultiSim, LTspice, Arduino IDE
- VLSI Competencies:** RTL Design, Digital Design, FPGA Design, High-Level Synthesis (HLS), Static Timing Analysis (STA), VLSI Design Flow (RTL-to-GDS), Physical Design
- Microfabrication:** Substrate cleaning, Thin film deposition, Lithography, Etching, Material characterization
- FPGA Platforms:** Nexys A7, ZedBoard, Spartan-6, Tang Nano 20K, ZCU104

WORK EXPERIENCE

Junior Research Fellow

Jul 2026 – Present

Sixth Sense Lab, IISER Bhopal, Madhya Pradesh, India

- Passionately working as JRF in the field of Semiconductor Micro/Nano Device Fabrication, focusing on the design, process development, and characterization of advanced semiconductor structures.
- My work involves hands-on experience with fabrication techniques, thin film deposition, lithography, and material characterization, aimed at developing efficient and scalable device architectures for semiconductor technology.

Project Intern – FPGA-Accelerated Bearing Fault Classifier

Jan 2026 – Present

SWAN Lab, IIT Kharagpur, West Bengal, India

- Architected a fully pipelined DSP system in Verilog, integrating an SPI-interfaced 16-bit ADC, dual-port circular BRAM, a time and Frequency domain pipeline delivering approximately 10x faster inference than an ESP32 implementation.
- Developed a Python feature extractor, and trained a Depth-10 scikit-learn decision tree achieving 81.83% multi-class fault detection accuracy; automatically translated the trained tree into synthesizable Verilog as a registered if-else chain, validated end-to-end from raw ADC samples to fault classification via an integration testbench.

Research Intern – Object Detection for ADAS

Jun 2025 – Jan 2026

Model Engineering College, Kochi | Chip to Startup (C2S) Program, MeitY, Govt. of India

- Designed and validated an object detection algorithm for ADAS, moving from Python algorithm testing to Verilog implementation on a ZCU104 FPGA using VITIS AI workflow, which produced a functional hardware prototype.
- Evaluated multiple detection models in Python and selected YOLOv5 and then Quantized the model using Vitis AI, achieving 80% reduction in model size, and generated a deployable XModel for execution on the ZCU104 FPGA.

PROJECTS

16x16 Systolic PE Array for Convolution Operations | [GitHub Link](#)

Sep 2025 – Dec 2025

Tools: Xilinx Vivado, Python | HDL: Verilog | Protocol: AXI4-Stream | FPGA: Nexys A7

- Executed a parameterized 16x16 Output-Stationary Systolic Array in Verilog for 2D convolution on FPGA, achieving 100% functional coverage against a cycle-accurate Python golden reference with zero numerical error across all 16 PEs.
- Conducted multi-precision hardware analysis across Q8.8, Q12.4, Q12.8, and Q12.12 fixed-point formats; identified Q8.8 as the optimal format at 0.119 W, and Q12.12 pipelining increased DSP utilization by 100% and on-chip power by 32% relative to Q8.8.
- Deployed Laplacian edge detection on a 640x640 KITTI image using the systolic array, decomposed the input into overlapping 6x6 tiles, streamed through Line Buffers for temporal alignment, and reconstructed the final 638x638 feature map from stitched 16x16 output blocks.

Pattern Detection Using CNN in Verilog | [GitHub Link](#)

Feb 2025 – Apr 2025

Tools: Xilinx Vivado | HDL: Verilog

- Implemented a 4-stage pipelined CNN accelerator in Verilog HDL for 8×8 pattern detection within a 128×128 grayscale image, integrating Laplacian edge enhancement, pattern-matching convolution, max pooling, and threshold-based detection as dedicated RTL modules.
- Designed a convolution stage using a parallel multiplier array with a hierarchical adder tree, compressing 36 partial products, achieving a maximum operating frequency of 155.67 MHz, and full image inference in 242 microseconds.

Image Enhancement on OCT Images Using FPGA | [GitHub Link](#)

Nov 2024 – Jan 2025

Tools: Xilinx Vivado, MATLAB | HDL: Verilog | FPGA: Artix-7

- Applied negative transformation, brightness control, and adaptive thresholding in Verilog for retinal OCT image enhancement on an Artix-7 FPGA.
- Simulated the Sobel edge detection algorithm to identify retinal layer boundaries and flag potential abnormalities.

WORKSHOPS & TRAINING

- **Summer School on Semiconductor Technology & Microfabrication – CeNSE, IISc Bangalore** | [Certificate](#)
Completed hands-on training in wafer processing, photolithography, doping, etching, thin-film deposition, and cleanroom protocols.
- **FPGA Design Trainee Program (C2S, MeitY) – Model Engineering College, Kochi** | [Certificate](#)
Built a Verilog-based FPGA Digital Payment System with core UPI functionalities.
1st Place in FPGA Design Contest, C2S Program, Model Engineering College, Kochi.
- **High-Level Synthesis for FPGA – Udemy**
Transformed combinational and sequential circuits in C++ using Vivado HLS, generated RTL IP cores, and verified designs on FPGA hardware.

CERTIFICATIONS

- System Design Through Verilog – NPTEL, IIT Guwahati | Score: 91% | Rank: Top 5% | [Certificate](#)
- VLSI Design Flow RTL to GDS – NPTEL, IIIT Delhi | Score: 88% | Rank: Top 5% | [Certificate](#)
- VLSI Physical Design – NPTEL, IIT Roorkee | Score: 83% | Rank: Top 5% | [Certificate](#)
- Electronic System Design and PCB Design – NPTEL, IIT Madras | Score: 85% | Rank: Top 5% | [Certificate](#)
- Verification Methodology – Maven Silicon | [Certificate](#)